

Rajat Kumar Saw

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Education

Indian Institute of Technology(Indian School of Mines) Dhanbad

Bachelor of Technology in Mining Machinery (Mechanical) Engineering

Expected May 2027

Current GPA: 8.83/10.0

Experience

Accenture

May 2026 – July 2026

Software Engineer Intern (AEH) – PySpark, Databricks, SQL, Azure

Bengaluru, India

- Built scalable ETL pipelines using **PySpark**, **Databricks**, **SQL**, and **Azure**, processing **10M+ records** while optimizing Spark transformations, partitioning, and joins to reduce end-to-end pipeline execution time by **35%**.
- Designed and implemented the **Medallion Architecture (Bronze, Silver, Gold)** using **Delta Lake**, improving data quality by **30%** and delivering reliable, analytics-ready datasets for downstream reporting and business intelligence.
- Architected optimized SQL queries and reusable PySpark workflows for data cleansing, validation, and aggregation, enabling **10+ business dashboards** with **99.9% data accuracy** while following production-grade data engineering best practices.

Projects

TermiNet – Web-Based Ephemeral Shell Access Platform | *Flask, Kubernetes, Docker, MySQL, Python*

- Developed a secure web-based shell interface using isolated Kubernetes pods, supporting **100+ concurrent users** for real-time Linux command execution.
- Orchestrated **on-demand pod creation** per user with enforced resource quotas and automatic teardown.
- Built a **Flask API** to manage user sessions, pod lifecycle, and command I/O through the **Kubernetes stream exec API**.

CryptTalk – Secure E2EE Chat | *React, Express.js, MongoDB, Socket.IO, Material UI, DHKE, AES*

- Developed a real-time chat application supporting **1-on-1 and group conversations** with a seamless and responsive user experience.
- Designed a **modern, responsive UI with Material UI** and prepared an **optimized paging system** for efficient message retrieval, minimizing server load and ensuring smooth performance at scale.
- Engineered **end-to-end encryption (E2EE)** with **Diffie-Hellman Key Exchange (DHKE)** for secure key agreement and **AES-256** for robust message encryption.

Technical Skills

- Languages:** C, C++, Python, JavaScript, HTML, CSS, SQL, Shell Scripting, Assembly
- Programming:** Competitive Programming, Data Structures & Algorithms (DSA), Object-Oriented Programming
- Web Technologies:** React, Express, Node.js, Flask, MongoDB, Git, GitHub, Docker, Kubernetes, Microsoft Azure
- Cybersecurity:** Web Exploitation, Cryptography, Binary Exploitation, Networking
- Tools:** Wireshark, Ghidra, Nmap, Burp Suite, Linux

Achievements

- Secured **2nd place** in ChronosCTF by IIT Mandi and ranked in the **Top 20** at IIT-BHU Codefest CTF.
- Successfully organized **Pearl CTF**, a **global cybersecurity competition** with participants from **28+ countries**.
- Solved **850+ DSA problems** on platforms like Codeforces, CodeChef and LeetCode.
- Achieved peak ratings of **1360+** on **Codeforces** and **1750+** on **LeetCode**.

Social Engagements

- Core Member:** CyberLabs- InfoSec Division, IIT Dhanbad - Conducted **10+ workshops** on cybersecurity.
- Event Organizer:** Prakash'25 - Annual Day of KARTAVYA – Led **event planning, logistics, and execution**, ensuring seamless operations and impactful engagement.
- Volunteer:** at KARTAVYA - NGO run by students of IIT Dhanbad to educate **underprivileged children**.